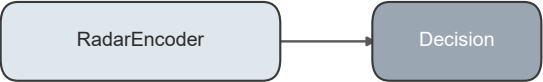


(a) Four architectures compared

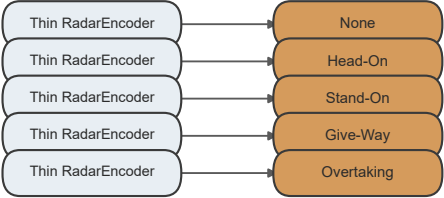
Single network

369K · one perception, one decision



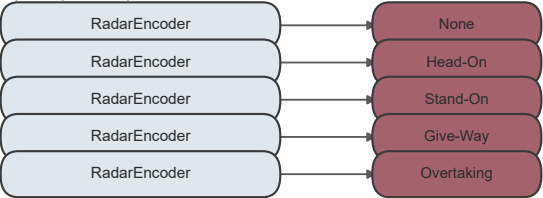
Separate experts, thin

363K · perception duplicated x5, each narrowed



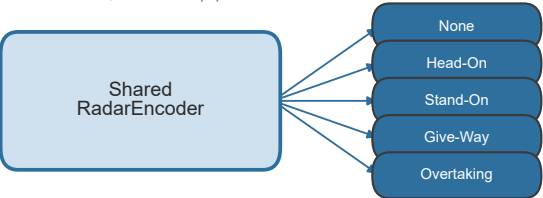
Separate experts, full

1.83M · perception duplicated x5, full width



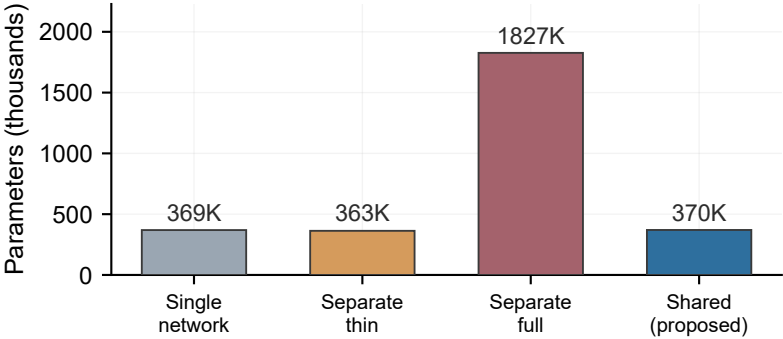
Shared perception (proposed)

370K · shared trunk, residual $\Delta\mu$ per situation

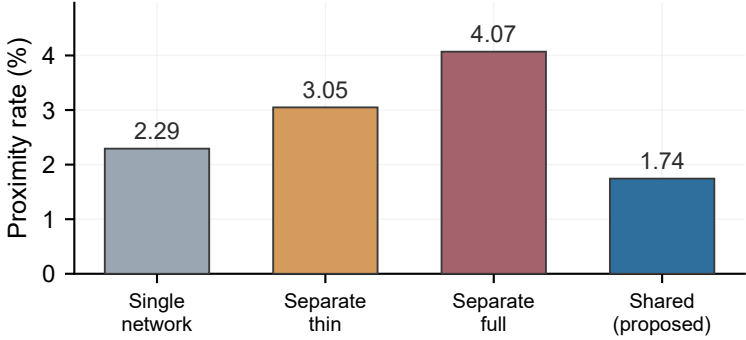


Radar perception is ~89% of single-network parameters.
The same expert decomposition is applied to MessageActor, ControlActor and Critic.

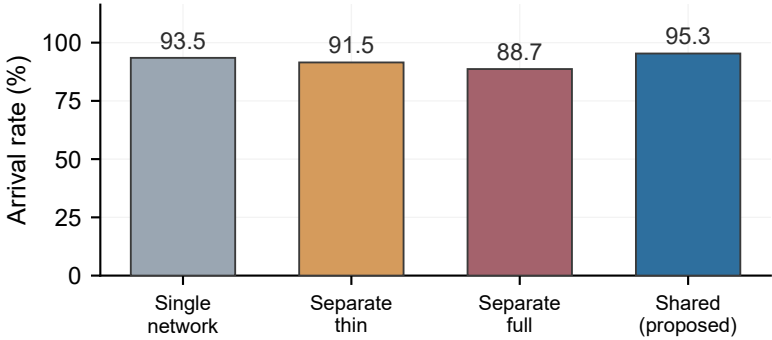
(b) Parameter count



(c) Proximity rate



(d) Arrival rate



Residual shared hub (fc2- μ shared, per-situation $\Delta\mu$, $\Delta\mu$ L2) vs single / thin / full, 3 seeds. Proposed hub wins arrival and proximity; parameter count matches the single network. Thin and full still duplicate the radar encoder and lose. THIN/THICK bars are v11 freeze (not residual).